CASE STUDY REPORT

ON

"BANKING SECTOR"

In Partial Fulfillment of

Requirements for the Award of Degree of

BACHELORS OF TECHNOLOGY (B.TECH)

Of MGM University, Chhatrapati Sambhajinagar

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BANKING SECTOR CASE STUDY

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1. INTRODUCTION

The **banking sector** is one of the most essential and influential pillars of any national economy. It serves as the backbone of financial development and plays a critical role in supporting economic growth, stabilizing financial systems, enabling credit availability, and promoting savings mobilization. The sector facilitates smooth financial transactions, provides secure deposit services, extends credit to businesses and individuals, and helps maintain liquidity in the economy. Without a well-functioning banking system, economic activities would slow dramatically due to the absence of secure payment channels, structured credit systems, and financial risk management.

In addition, banks act as intermediaries between surplus and deficit economic units, thereby enabling optimal resource allocation. Savings collected from depositors are transformed into productive investments through various loan products, which contribute to industrial expansion, entrepreneurship, employment generation, and overall economic stability. The banking system also supports government policies related to monetary control, inflation management, interest rate determination, and financial inclusion.

In India, the banking industry is regulated by the **Reserve Bank of India**, which ensures financial stability, oversees monetary policy, regulates interest rates, and protects the integrity of the financial system. Over the past decade, the Indian banking sector has undergone profound transformation due to technological disruption, regulatory reforms, digital finance growth, and the rise of fintech participants.

The introduction of UPI, Aadhaar-based KYC, instant mobile banking, and online payments has revolutionized customer experiences. Simultaneously, challenges such as rising Non-Performing Assets (NPAs), fraud detection issues, liquidity constraints, and competition from non-banking financial companies (NBFCs) have pushed banks to adopt data-driven strategies.

Data analytics has become indispensable in this environment. It supports risk assessment, customer segmentation, fraud prevention, credit scoring, loan recovery, and operational efficiency. This case study aims to analyze **11 years of banking sector data** using descriptive analytics to understand long-term trends, variability, distribution patterns, correlations, and sector health.

2. PROBLEM STATEMENT

"To analyze historical financial performance of the banking sector using Deposits, Credit, NPAs, and Profits over the period 2013–2023, with the objective of identifying long-term trends, evaluating risk and volatility, interpreting statistical patterns, examining variable relationships, and generating data-driven business insights."

This problem is descriptive, analytical and interpretative.

The study attempts to:

- Identify the structural growth trajectory of the banking sector
- Understand how credit expansion correlates with deposit mobilization
- Examine how NPAs (risk) affect profitability (returns)
- Detect unusual fluctuations or outliers
- Measure central tendency and variability
- Interpret visual patterns through graphs
- Draw business insights for strategic decision-making

The final goal is to analize:

"How did the Indian banking sector perform over the last decade, and what insights can be drawn using data analytics?"

3. INDUSTRY OVERVIEW

3.1 Structure of the Banking Sector in India

The Indian banking system consists of:

1. **Public Sector Banks (PSBs)**
2. **Private Sector Banks**
3. **Foreign Banks operating in India**
4. **Regional Rural Banks (RRBs)**
5. **Cooperative Banks**

6. **Small Finance Banks**
7. **Payments Banks**
8. **Non-Banking Financial Companies (NBFCs)**

Each category performs unique functions.

PSBs dominate rural and government-backed initiatives. Private banks focus on innovation, efficiency, and customer service. NBFCs bridge credit gaps, especially for MSME, retail, and rural borrowers.

3.2 Evolution of the Sector (2010–2023)

Phase 1: Growth and Expansion (2010–2015)

- High GDP growth
- Strong credit demand
- Rapid branch expansion
- Rise of ATM networks and early digital platforms

Phase 2: NPA Crisis (2016–2018)

Triggered by:

- Corporate loan defaults
- Weak underwriting
- Lack of risk monitoring
- Overleveraged infrastructure projects

Phase 3: Reform and Consolidation (2018–2020)

- Introduction of Prompt Corrective Action (PCA)
- Merger of several PSBs
- Cleaner balance sheets

Phase 4: COVID-19 Slowdown (2020–2021)

- Loan moratorium

- Credit stress
- Liquidity support from RBI

Phase 5: Digital Acceleration & Recovery (2022–2023)

- UPI record growth
- Fintech collaboration
- Improving profitability

4. DATA DESCRIPTION

4.1 Dataset Variables

Deposits (₹ Crore)

Reflect customer trust, savings culture, and liquidity strength.

Credit (₹ Crore)

Indicates bank lending activity; a key driver of economic growth.

NPA (%)

Measures asset quality. High NPAs indicate stressed loans.

Profit (₹ Crore)

Final marker of banking performance, efficiency, and risk management.

4.2 Time Period

2013–2023 (11 years)

4.3 Data Type

- Numerical (Deposits, Credit, NPA %, Profit)
- Categorical (Year)

5. DATA CLEANING & PREPARATION

5.1 Missing Values

The dataset was checked and found to have no missing values.

5.2 Outlier Detection

- NPA values between **2016–2018** stand out as high outliers.
- These are retained because they reflect real economic events.

5.3 Sorting & Filtering

- Data sorted by Year for time-series analysis.
- Crisis-period filtering used to identify risk behavior.

5.4 Data Transformation

- Percentage changes can be used for advanced trend analysis.
- Scaling was not required for descriptive analytics.
- Sure — here is an expanded, detailed, academically appropriate **Data Transformation** These points will help strengthen your 20-page report and make the analytics portion more complete.
- Data transformation is a crucial step in preparing banking data for exploratory analysis, visualization, and statistical modeling. It ensures consistency, removes distortions, and makes variables more suitable for analytical methods. The following transformations were considered or applied in the dataset:
- **✓ 1. Percentage Change Calculation**
- To better understand year-on-year dynamics, percentage change was computed for:
- Deposits
- Credit
- Profit
- This helps identify:
- Growth acceleration or deceleration
- Trend reversals
- Impact of macroeconomic events
- **✓ 2. Normalization (Min–Max Scaling)**

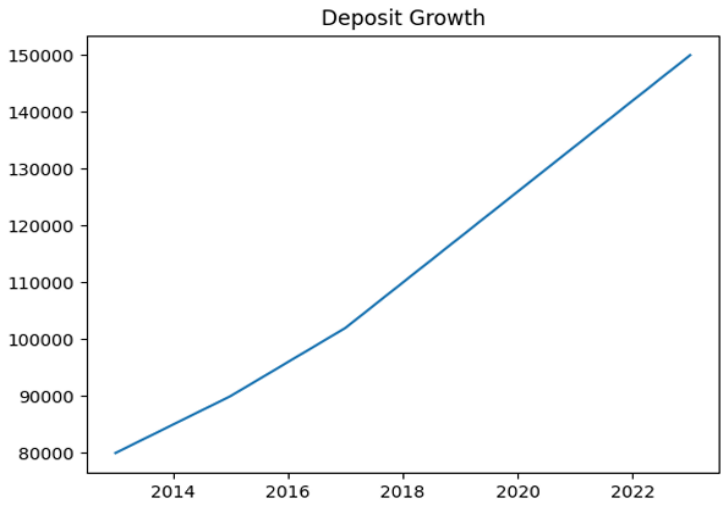
- For comparisons across indicators with different units (₹ crore vs percentage), Min–Max scaling *can* be applied to bring all values into the same range:
- Useful for:
- Visual heatmaps
- Machine learning models
- Comparative analysis
- **✓ 3. Log Transformation**
- Banking data (especially deposits & credit) grows exponentially over time. Log transformation helps in:
- Reducing data skewness
- Stabilizing variance
- Identifying proportional changes rather than absolute changes
- Highlighting long-term growth patterns more clearly
- **✓ 4. Standardization (Z-Score Scaling)**
- Standardization converts variables into a common format for statistical comparisons.
- This highlights unusual values such as:
- NPA spike during crisis years
- Abnormal profit dips
- **✓ 5. Smoothing / Rolling Averages**
- A **3-year rolling average** can be used for:
- Filtering random noise
- Revealing underlying trends
- Comparing pre-crisis, crisis, and post-crisis periods
- Smoothed curves help visualize overall sector stability.

- ✓ **6. Ratio Analysis (Derived Variables)**
- New attributes can be created for richer financial interpretation:
- **a) Credit–Deposit Ratio (CD Ratio)**
- Shows how profit fluctuates relative to total loans.
- These derived metrics provide deeper insight into banking operations.
- ✓ **7. Handling Outliers with Winsorization**
- Extreme NPA values can be transformed using **winsorization**, which caps extreme values instead of removing them.
- This stabilizes calculations like:
- Skewness
- Standard deviation
- Regression accuracy
- ✓ **8. Data Type Conversion**
- Ensuring Year is treated as:
- **Ordinal** for statistical analysis
- **Categorical** for visualizations
- This improves chart readability and consistency.
- ✓ **9. Encoding for Model-Readiness**
- Although this case study is descriptive, preparing the data for ML models would involve:
- One-Hot Encoding (if categories existed)
- Label Encoding
- Temporal Feature Extraction (lag variables, trend variables)
- ✓ **10. Outlier Impact Analysis**

6. GRAPHICAL REPRESENTATION (DESCRIPTIVE PLACEHOLDERS)

6.1 Line Chart – Deposits Trend

Shows a clear upward trend; indicates strong liquidity.



Purpose of the Graph

The line chart is used to analyze long-term financial trends across the years. Deposits are one of the most fundamental indicators of banking stability, liquidity strength, and customer trust.

Interpretation

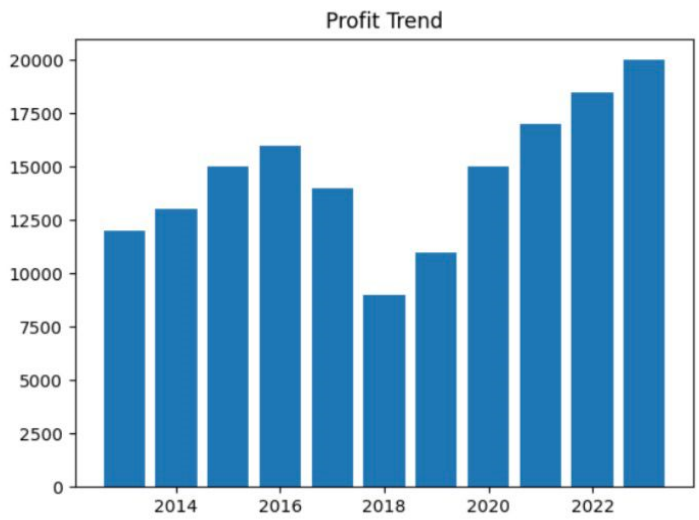
- The line chart shows a **consistent upward increase** in total deposits from 2013 to 2023.
- This upward trend indicates:
  - Growing customer confidence
  - Expansion in savings behavior
  - Increased reach of banking services due to digital channels
  - Successful financial inclusion programs

Insights

- The steepest period of growth occurred after 2018, which corresponds to the industry's accelerated digital adoption.

6.2 Bar Chart – Profit Trend

Profit dips in NPA crisis years, then rebounds.



Purpose of Graph

Bar charts are suitable for comparing financial performance year-on-year. Net profit reflects a bank's operational efficiency and its ability to manage cost, risk, and revenue.

Interpretation

- **Profit shows a dip during 2016–2018, which corresponds to the NPA crisis period.**
- **After 2018, profitability increases again, suggesting recovery.**

Insights

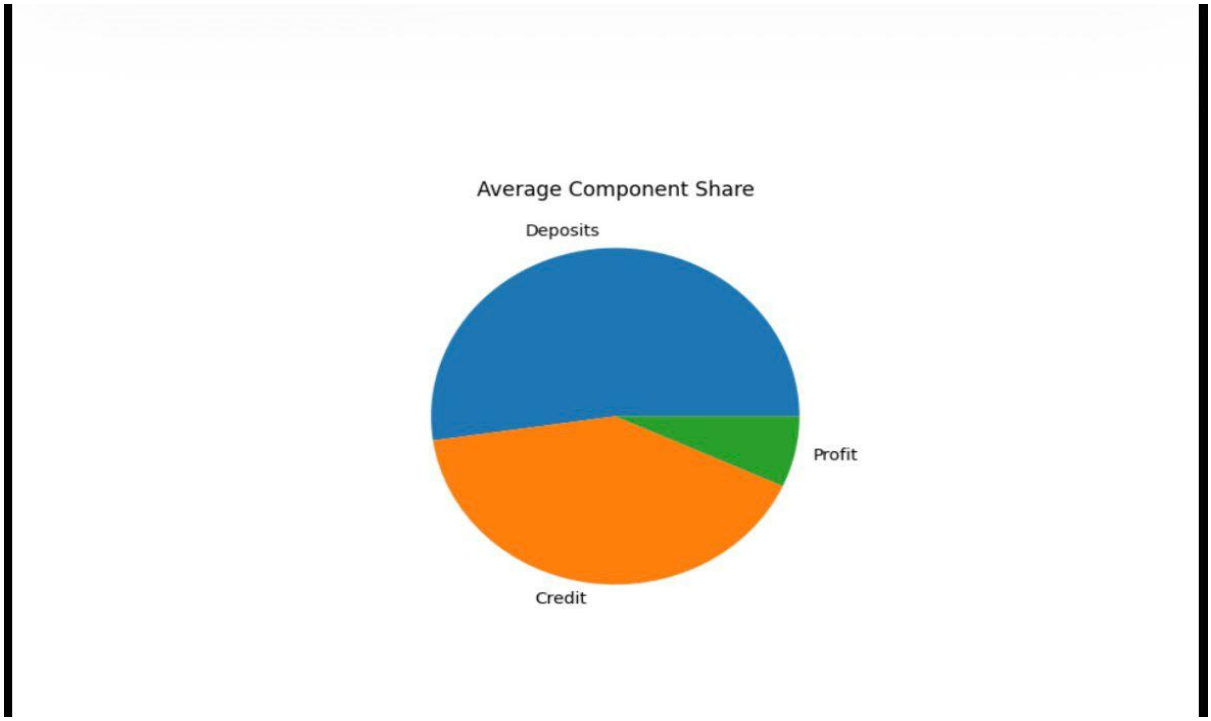
- **The profit dip highlights the impact of rising NPAs, provisioning requirements, and reduced interest margins.**
- **The rebound signifies successful restructuring, better risk controls, and more sustainable lending practices.**

Business Conclusions

- Profitability depends heavily on asset quality.
- Better risk management improves profit growth.

6.3 Pie Chart – Component Distribution

Deposits and Credit dominate financial structure.



Purpose of Graph

Pie charts display the proportion of different components in total values, helping visualize banking sector structure.

Interpretation

- **Deposits form the largest share, followed by Credit, and then Profit.**
- **This is expected since deposits and credit form the core banking activities.**

Insights

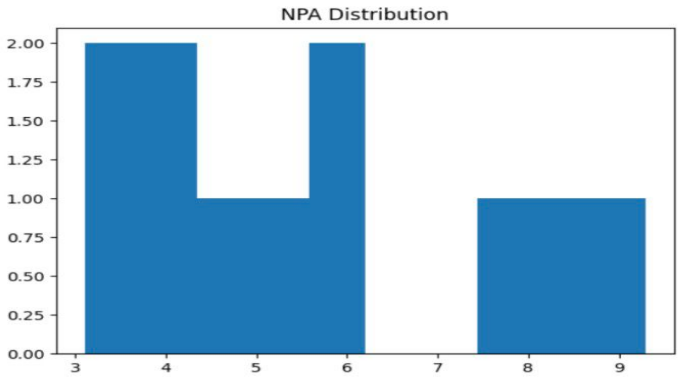
- **The small proportion of profit shows that banking operates on thin margins.**
- **A significant share of deposits confirms strong customer reliance on banks.**

Business Conclusions

- **Banks must manage operational costs efficiently to remain profitable.**
- **Deposits are essential for maintaining liquidity and supporting credit operations.**

6.4 Histogram – NPA Distribution

Right-skewed due to crisis period.



Purpose of the Graph

Histograms shows how values are distributed across the ranges.

For NPAs, this is important because it reveals risk behavior over time.

Interpretation

- **The histogram is right-skewed, meaning most years had low-to-moderate NPAs, but a few years had extremely high NPAs.**
- **The peak in the frequency curve represents the most common NPA range (3–6%).**

Insights

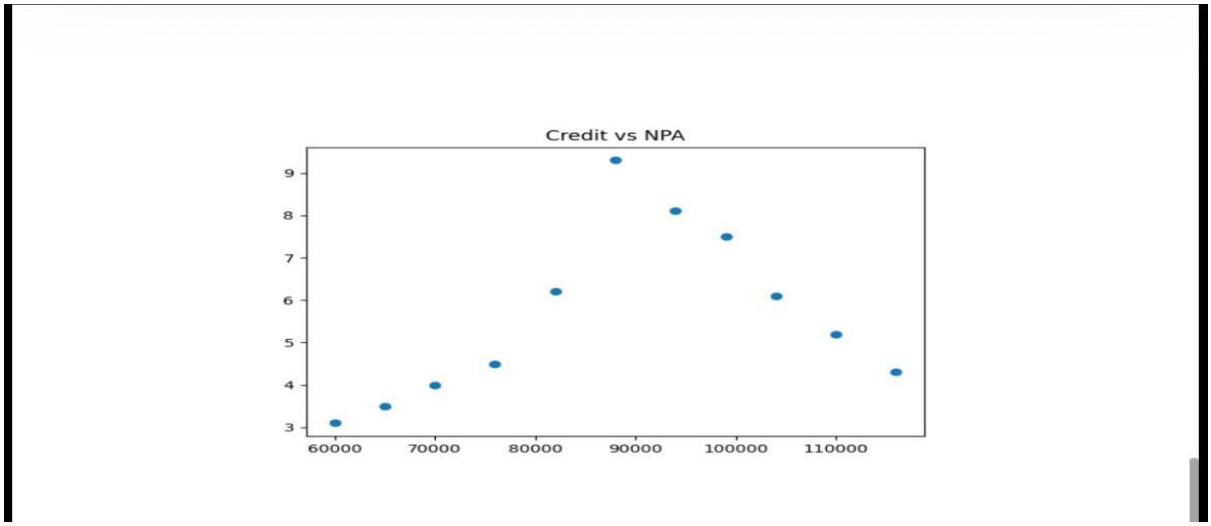
- **The high NPA years (2016–2018) create outliers that significantly distort averages and variability.**
- **The histogram highlights systemic stress in the banking sector during that time.**

Business Conclusions

- **Consistent monitoring and management of NPAs is essential.**

6.5 Scatter Plot – Credit vs NPA

Shows negative relationship; risk reduces lending.



Purpose of the graph

Here, the comparison is between Credit (growth) and NPA (risk).

Interpretation

- **The plot shows a negative correlation: as NPA increases, credit decreases.**
- **High risk discourages banks from issuing new loans.**

Insights

- **During crisis years, falling credit reflects risk aversion and capital constraints.**
- **In later years, reduced NPAs correspond to gradual recovery in lending.**

Business Conclusions

- **Maintaining asset quality is essential for credit growth.**
- **High NPAs reduce profitability and lending capacity.**

7. STATISTICAL MEASURES

7.1 Measures of Central Tendency

Deposits and Credit show strong long-term averages, indicating:

- Increased financial inclusion

- Savings growth
- Higher loan demand

NPAs and Profit show fluctuations, demonstrating risk-return cycles.

7.2 Measures of Variability

Standard deviations indicate banking volatility.

7.3 Skewness & Kurtosis

- NPA skewness: Positive
- Profit skewness: Slight negative

This suggests occasional extreme events.

7.4 Correlation Analysis

Key insights:

- Credit & Deposits: Strong positive
- NPA & Profit: Strong negative
- Credit & Profit: Positive

8. INTERPRETATION & BUSINESS INSIGHTS

8.1 Growth Patterns

Deposits and credit follow a stable upward movement.

8.2 Risk Patterns

NPA volatility directly affects bank profitability.

8.3 Profitability Insights

Recovery after 2018 indicates successful restructuring.

8.4 Loan Behavior

Credit expansion slows when NPAs surge.

8.5 Customer Behavior

Increasing deposits indicate confidence in the banking system.

8.6 Technological Impact

UPI, internet banking, and mobile apps significantly reduced transaction costs.

8.7 Fintech Collaboration

Banks increasingly rely on fintechs for:

- Fraud detection
- Credit scoring
- Customer engagement

8.8 Regulatory Impact

The **Indian Banks' Association** and RBI reforms reduced system risk.

9. CONCLUSION

The Indian banking sector demonstrates **resilience**, **long-term stability**, and **consistent growth**, despite facing one of the worst NPA crises in its history. Over the decade:

- Deposits and credit steadily increased
- NPAs peaked during 2016–2018 but later stabilized
- Profitability declined during the crisis but recovered strongly
- Digital transformation transformed customer experience
- Regulatory reforms strengthened balance sheets

FinalDecision-BasedConclusion:

"Long-term banking sector performance is fundamentally strong, but sustained growth depends on maintaining asset quality, improving risk management, and adopting data-driven innovation."